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Part 1: Find the best AI Employee

1. List of your chosen 3 LLM's
    a. ChatGPT
    b. Google Gemini
    c. Claude AI
2. Craft 5 Interview Questions regarding web development.
    a. Craft a Simple application with HTML, CSS, and JS. (Snake Game, Minesweeper, etc)
    b. Explain the code for the following application from the 1st question
    c. Make a simple portfolio website using HTML, CSS and JS
    d. Include a working API for the following portfolio website
    e. What are the easiest technology stacks that can be used for web development? Explain it short and simple.
3. Report on the results of the chosen LLM from the interview question. (ChatGPT)
    a. Crafting Snake Game

        Accurate: Functional snake game with minor bugs (wall collision bug) but can't replay the game
        Clear: Good code and easy to understand
        Efficient: Fast response and very simple solution

    b. Explain the Code for Snake Game

        Accurate: Explained the code correctly from the code given
        Clear: Structured the code to multiple parts and easy to understand explanation
        Efficient: Fast response and no redundant explanation

    c. Portfolio Website

        Accurate: Portfolio website working with no errors. Lacks design
        Clear: The code was clear. However, no instructions were given for portfolio customization
        Efficient: Quick and simple respond. Very simple and efficient solution

    d. Including API

        Accurate: Random API integrated with no error.
        Clear: Code was very simple and instructions is very easy and clear
        Efficient: Quick response however API doesn't suit with the portfolio project thus less efficient

    e. Technology Stack for Web Development

        Accurate: Listed 5 web development stacks which are simple and beginner friendly
        Clear: Clear distinction and explanation
        Efficient: Fast response and correct
4. Report on the results of the chosen LLM from the interview question. (Google Gemini)
    a. Crafting Snake Game

        Accurate: Fully functioning, and simple design

Clear: Clean code structure and perfect commenting
Efficient: Clean solution with quick response

b. Explain the Code for Snake Game

Accurate: Explanation covered most of the snake game however missed some of the major components such as the canvas
Clear: Explanation was clear however it is not very detailed
Efficient: Fast response however not detailed, make it less efficient

c. Portfolio Website:

Accurate: Website worked perfectly with very aesthetic design
Clear: Documentation and code was clear with instructions for customization
Efficient: Very Efficient, very fast and very reliable

d. Including API

Accurate: Advanced API integration that suits with the portfolio website with no errors.
Clear: Instructions given and code are understandable and clear
Efficient: Very efficient and quick response

e. Technology Stack for Web Development

Accurate: Answer provided 3 different technology stack that are complete and more detailed
Clear: Clear explanation and gave scenarios
Efficient: Quick and simple response

5. Report on the results of the chosen LLM from the interview question. (Claude AI)

a. Crafting Snake Game

Accurate: Playable, have delay in control and a lot of extra features
Clear: Very complex coding and overwhelming explanation
Efficient: Unnecessary complex codes for unnecessary features, long response time

b. Explain the Code for Snake Game

Accurate: Correctly explain the code given. However, did not specify the delay in movement
Clear: Clean explanation structure but too much detail making it harder to understand
Efficient: Very long waiting time making it not efficient with multiple unnecessary explanations

c. Portfolio Website:

Accurate: The portfolio website works very well with very advanced features and complex design which are very good for a portfolio
Clear: Due to advance features and design, code become more complex and require more knowledge to understand
Efficient: Efficient to take the fully code work however might take some time for its response (more than 9 trials)

d. Including API

Accurate: Included an AI Chatbot, error on running as it is not working.
Clear: No documentation for the AI chat widget however, it separates and mention specific structure for the AI API implementation
Efficient: Not so efficient as it lacks function and took a lot of resources.

e. Technology Stack for Web Development

Accurate: Correctly explain technologies stack with very high detail even after asking it to make it simple
Clear: Explanation and detail is clear
Efficient: Fast response however not as efficient as it doesn't follow instructions

6. Based on your interviews, which LLM had the best responses. Explain your answer.

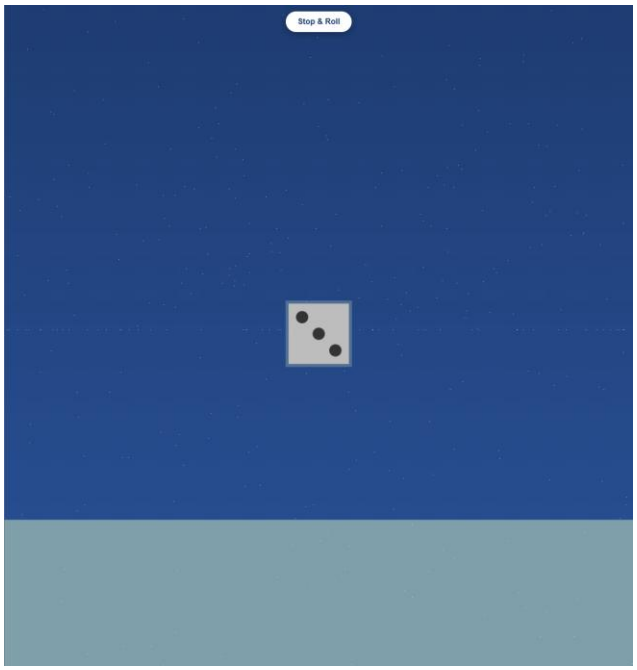
a. Google Gemini is the chosen LLM that had the best response as it gave the most consistent results that have high ACEs which are mostly accurate, provided clear code and instructions and shows high efficiency. ChatGPT is second as it provides clearer explanations and works very efficiently with slight accuracy. Claude AI is last as it is not as efficient as the other two even after showing high and complex designs and features.

Part 2: AI Guidance for Unknown Topics

Prompts Used

1. Create a 2D canvas and a bouncing 3D dice cube program using three.js in a html, css, and js program
2. Make the dice bouncier and make the dice bounce when it hit each wall or end of screen
3. Create a revised code where the background of the program is snowing with light blue as the floor and blue as the sky
4. Create a revised code to make the dice bouncier and change the sides from number to dots
5. Create a revised code to make the dice draggable by the user
6. Create a revised code where the dice is draggable wherever the user click such that the user doesn't have to click the dice to drag it
7. Create a revised code to add a button with label stop dice on the top of the screen where when pressed, the dot will stop in the middle of the screen, showing one side of the dice
8. Create a revised code where the stop button will pick a random side of the dice to show every time it is pressed

Github Link: djm7749/cmpsc263-ai-assignment

Prompt: https://gemini.google.com/share/b2a1295c4d0e

Screenshot: